



LOGICAL REASONING & ANALYTICAL SKILLS

QUESTION BANK

MODULE I: DATA ARRANGEMENTS

SHORT ANSWERS

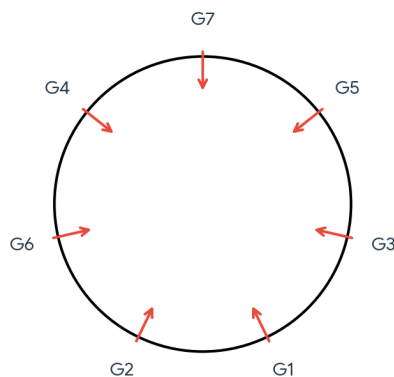
1) Seven girls G1, G2, G3, G4, G5, G6, and G7 are sitting around a circular table facing the center (not necessarily in the same order). G6 is second to the left of G1. G7 is second to the left of G6. G5 is third to the right of G2. G5 is not the immediate neighbour of G6, G4 is third to the right of G3. Who is second to the left of G7?

(A) G2

(B) G1

(C) G4

(D) G3



2) Seven people, A, B, C, D, E, F and G are sitting in a straight line, facing north. Only three people sit to the left of D. Only two people sit between D and A. B sits fifth to the right of C. G sits to the immediate right of E but somewhere to the left of A. Who sits to the extreme right?

(A) B

(B) A

(C) E

(D) C

1	2	3	4	5	6	7
C	E	G	D	F	B	A



3) Eight friends A, B, C, D, E, F, G and H are seated in a straight line and all of them are facing north, but not necessarily in the same order. B sits second to right of F. F sits at one of the extreme ends of the line. Only three people sit between B and E. C sits third to the left of D. Only two people sit between D and A. G is not an immediate neighbour of E. Who is sitting on the immediate left of B?

(A) F

(B) A

(C) C

(D) G

1	2	3	4	5	6	7	8
F	C	B	G	D	H	E	A

4) In a shop, there were 4 dolls of different heights A, B, C and D. D is neither as tall as A nor as short as C. B is shorter than D but taller than C. If Mani wants to purchase the tallest doll, which one should she purchase?

(a) Only A

(b) Only D

(c) Either A or D

(d) Either B or D

$$A > D > B > C$$

5) Five men A, B, C, D and E read a newspaper. The one who reads first gives it to C. The one who reads last had taken from A. E was not the first or last to read. There were two readers between B and A.

1	2	3	4	5
B	C	E	A	D

(I) B passed the newspaper to whom?

a) A

(b) C

(c) D

(d) E

(II) Who read the newspaper last?

(a) A

(b) B

(c) C

(d) D



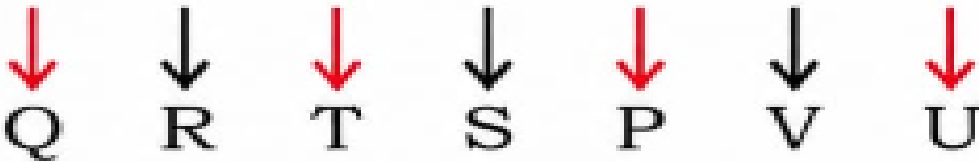
6. Seven girls P, Q, R, S, T, U, and V are sitting in facing the south (not necessarily in the same order). Q and R are immediate neighbours of each other. Q is at one of the extreme ends. P is fourth to the left of Q. U is not the immediate neighbour of either P or R. S is second to the right of V. Who is sitting on the immediate right of R?

(A) Q

(B) P

(C) T

(D) V





LONG ANSWERS

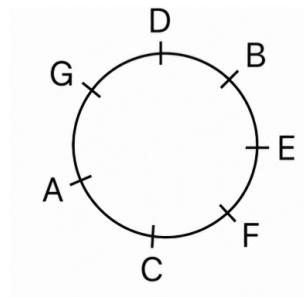
1. Read the following information carefully and answer the questions given below:

(i) A, B, C, D, E, F and G are sitting around a circle and are facing the centre.

(ii) G is second to the left of C, who is to the immediate left of F.

(iii) A is third to the left of E.

(iv) B is between D and E.



1. Which of the following is false?

(a) A is fourth to the right of E.

(b) G is to the immediate right of D.

(c) F is third to the right of D.

(d) B is to the immediate left of D.

2. Which of the following is true?

(a) C is fourth to the left of B.

(b) A is to the immediate right of G.

(c) D is second to the left of E.

(d) B is second to the right of G.

3. Which of the following pairs has the first person sitting to the immediate left of the second person?

(a) BE

(b) CA

(c) GD

(d) DG

4. Which of the following has the middle person sitting between the remaining two?

(a) FCE

(b) EFB

(c) DEB

(d) None of these

5. Which of the following is the position of F?

(a) Fourth to the right of D

(b) To the immediate left of C



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2. Study the given information carefully and answer the questions that follow:

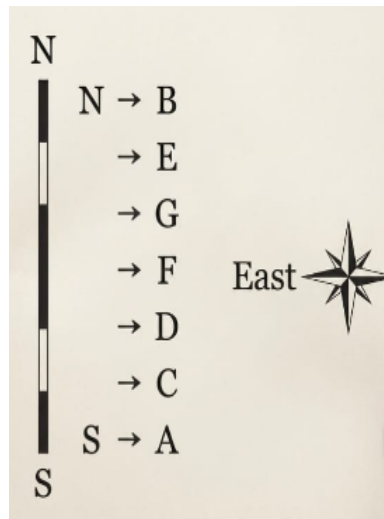
(i) A, B, C, D, E, F and G are sitting on a wall and all of them are facing east.

(ii) C is on the immediate right of D.

(iii) B is at an extreme end and has E as his neighbour.

(iv) G is between E and F.

(v) D is sitting third from the south end.



1. Who is sitting to the right of E?

(a) A

(b) C

(c) D

(d) G

2. Which of the following pairs of people are sitting at the extreme ends?

(a) AB

(b) AE

(c) CB

(d) FB

3. Name the person who should change place with C such that he gets the third place from the north end.

(a) E

(b) F

(c) G

(d) D

4. Immediately between which of the following pairs of people is D sitting?

(a) AC

(b) AF

(c) CE

(d) CF



5. Which of the conditions (i) to (v) given above is not required to find out the place in which A is sitting?

- (a) (i) (b) (ii) (c) (iii) (d) All are required

3. Study the following information carefully to answer the questions given below:

P, Q, R, S, T, U and V are seven professors. Each one teaches a different subject from Physics, Chemistry, Biology, English, Mathematics, Economics and Geography, not necessarily in the same order. Each of them teaches one day each on seven days of the week from Monday to Sunday not necessarily in the same order. R teaches Biology on Friday. Q teaches Mathematics on the previous day of the day on which the professor teaches Physics. V teaches on Sunday but does not teach Chemistry or English. S teaches Economics on the previous day on which U teaches. P teaches Geography on Tuesday. T does not teach English.

Professor	P	Q	R	S	T	U	V
Day	Tues-day	Satur-day	Friday	Wednes-day-Monday	Thurs-day	Sunday	Sun-day
Subject	Geog-raphy	Math-ematics	Biology	Econom-ics	Chem-istry	Eng-lish	Phys-ics

1. Which subject does T teach?

- (a) Chemistry (b) Physics (c) Either Chemistry or Physics (d) Data inadequate

2. Which subject does U teach?

- (a) Physics (b) English (c) Either Physics or English (d) Data inadequate

3. On which day does T teach?

- (a) Monday (b) Wednesday (c) Thursday (d) Data inadequate



4. On which day does U teach?

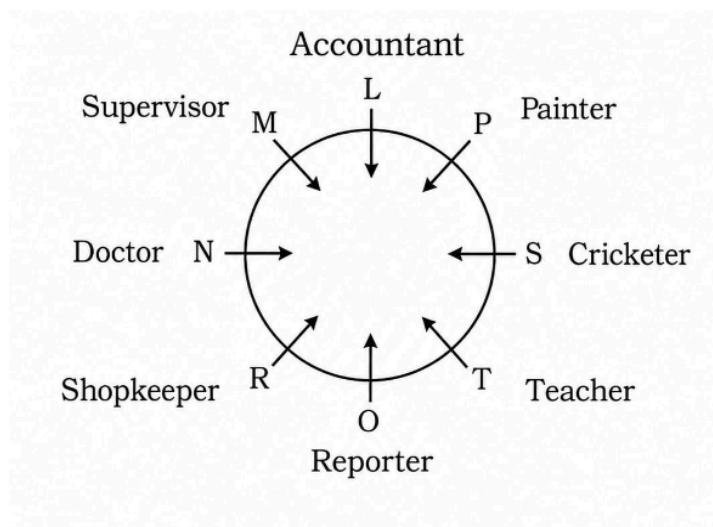
- (a) Wednesday (b) Thursday (c) Wednesday or Thursday (d) Data inadequate

5. Which subject does V teach?

- (a) Chemistry (b) English (c) Biology (d) Physics

4) Study the following information carefully and answer the questions given below:

Eight people- S, R, N, L, M, T, O and P are sitting in a circle facing the center. All eight belong to different professions- reporter, doctor, cricketer, teacher, accountant, shopkeeper, painter and supervisor. They are not necessarily seated in the mentioned order. M is sitting third to the left of O. The doctor is to the immediate right of M and M is not a reporter. R is sitting fourth to the right of P. Neither R nor P is an immediate neighbour of M. T is a teacher and is sitting third to the right of the doctor. The shopkeeper is sitting second to the left of the teacher. The painter is sitting second to the left of M. S the cricketer is sitting exactly between T and P. The accountant is sitting second to the right of the cricketer. N is sitting third to the left of T.





(1) Who among the following is a supervisor?

- (A) S (B) N (C) L (D) M

(2) Who among the following is a reporter?

- (A) S (B) O (C) L (D) N

(3) Who is sitting on the immediate left of the shopkeeper?

- (A) L (B) P (C) N (D) S

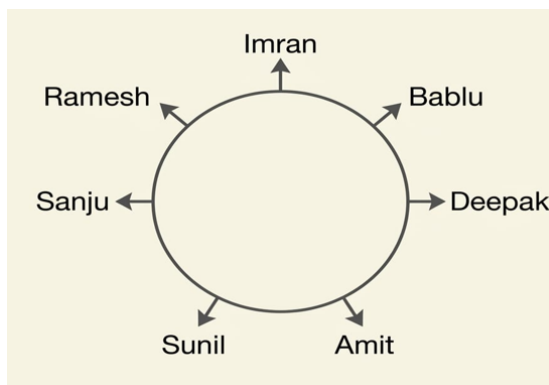
(4) How many people sit between teacher and supervisor?

- (A) Only 1 (B) Only 2 (C) Only 3 (D) Only 4

(5) Who are immediate neighbours of accountant?

- (A) N & S (B) M & P (C) S & P (D) N & M

5. Amit, Bablu, Ramesh, Deepak, Imran, Sunil, and Sanju are sitting around a circular table facing opposite the center. Amit does not sit adjacent to Imran and Sanju Bablu does not sit adjacent to Amit. Sanju sits second to the left of Imran. Ramesh sits fourth to the left of Amit. Deepak and Bablu sit adjacent to each other



1) Who sits to the immediate right of Amit?

- (A) Bablu (B) Sunil (C) Ramesh (D) Sanju



2. Who sits second to the left of Imran?

- A) Amit **B) Sanju** C) Deepak D) Sunil

3. Who sits fourth to the left of Amit?

- A) Bablu **B) Ramesh** C) Imran D) Sunil

4. Which of the following pair sits adjacent to each other?

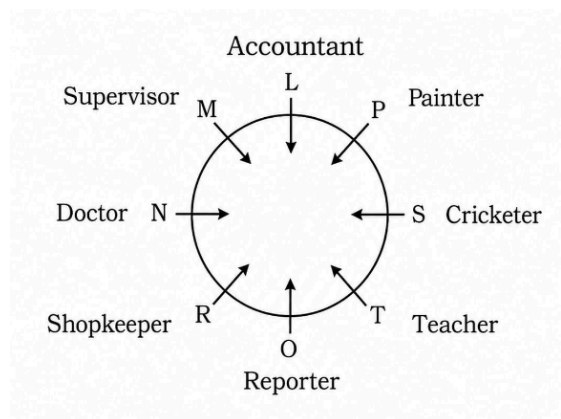
- A) Amit and Bablu **B) Deepak and Bablu** C) Amit and Imran D) Sanju and Amit

5. Amit is not adjacent to which of the following?

- A) Imran and Sanju** B) Deepak and Bablu C) Ramesh and Sunil D) Sunil and Deepak

6) Study the following information carefully and answer the questions given below:

Eight people- S, R, N, L, M, T, O and P are sitting in a circle facing the center. All eight belong to different professions- reporter, doctor, cricketer, teacher, accountant, shopkeeper, painter and supervisor. They are not necessarily seated in the mentioned order. M is sitting third to the left of O. The doctor is to the immediate right of M and M is not a reporter. R is sitting fourth to the right of P. Neither R nor P is an immediate neighbour of M. T is a teacher and is sitting third to the right of the doctor. The shopkeeper is sitting second to the left of the teacher. The painter is sitting second to the left of M. S the cricketer is sitting exactly between T and P. The accountant is sitting second to the right of the cricketer. N is sitting third to the left of T.



(1) Who among the following is a supervisor?



(2) Who among the following is a reporter?

- (A) S (B) O (C) L (D) N

(3) Who is sitting on the immediate left of the shopkeeper?

- (A) L (B) P (C) N (D) S

(4) How many people sit between teacher and supervisor?

- (A) Only 1 (B) Only 2 (C) Only 3 (D) Only 4

(5) Who are immediate neighbours of accountant?

- (A) N & S (B) M & P (C) S & P (D) N & M

MODULE 2

DATA SUFFICIENCY

Short Answers

1) What is the colour of the fresh grass?

I. Blue is called green, red is called orange, orange is called yellow,

II. Yellow is called white, white is called black, green is called brown and brown is called purple.

Sol: The colour of fresh grass is 'green' and as given in II, 'green' is called 'brown'. So, the colour of fresh grass is 'brown'.

1) If the data under statement I alone is sufficient to answer the question.

2) If the data under statement II alone is sufficient to answer the question.



3) If the data under both the statements (I and II) together are necessary to answer the question.

4) If the question cannot be answered even by using both the statements together.

2) What does 'nip' stand for in a code language?

I. In the code language, 'that is very beautiful' is written as 'se nip sre num'.

II. In the same code language, 'my house is beautiful' is written as 'nip sto sre tip'.

Sol: In I and II, the common codes are 'nip' and 'sre' and the common words are 'is' and 'beautiful'. So, 'nip' and 'sre' are the codes for 'is' and 'beautiful'. But, the exact word for 'nip' cannot be found out.

- 1) If the data under statement I alone is sufficient to answer the question.
- 2) If the data under statement II alone is sufficient to answer the question.
- 3) If the data under both the statements (I and II) together are necessary to answer the question.
- 4) If the question cannot be answered even by using both the statements together.

3) What is the percentage of profit made by selling milk?

I) The selling price of the milk was Rs. 25 per litre.

II) The milk sold at cost price after adding 250ml water for litre.

$$\text{from 2} \Rightarrow 250\text{ml/L}$$
$$\Rightarrow \frac{250}{1000} \times 100 = 25\%$$

- 1) If the data under statement I alone is sufficient to answer the question.
- 2) If the data under statement II alone is sufficient to answer the question.



3) If the data under both the statements (I and II) together are necessary to answer the question.

4) If the question cannot be answered even by using both the statements together.

MODULE 2

DATA SUFFICIENCY

Long Answers

1) What is the area of the rectangular plot?

I. The perimeter of the plot is 208 metres.

II. The length is more than the breadth by 4 metres.

from I + II

$$\begin{aligned}\text{let breadth} &= x \\ \text{length} &= (x + 4) \\ \text{Perimeter} &= 2(a+b) = \cancel{2(x+4)} \\ &= 2(l+b) \\ &= 2(x + x + 4) \\ &= 4x + 8 = 208 \\ x &= 50 \\ B &= 50 \quad L = 54 \\ \text{Area} &= l \times b = 54 \times 50 \\ &= 2700 \text{ m}^2\end{aligned}$$

1) If the data under statement I alone is sufficient to answer the question.



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2) If the data under statement II alone is sufficient to answer the question.

3) If the data under both the statements (I and II) together are necessary to answer the question.

4) If the question cannot be answered even by using both the statements together.

2) Total money with Naresh and Ajay is 28 per cent of that with Usman. How much money is Ajay having?

I. Usman has got Rs 75000.

II. The ratio of money of Naresh to money held by Ajay is 1: 3.

From I, total money with Naresh and Ajay =
 $\text{₹} \left(\frac{28}{100} \times 75000 \right) = \text{Rs. } 21000.$
Thus, from II, money with Ajay
 $= \text{₹} \left(\frac{3}{4} \times 21000 \right) = \text{Rs. } 15750.$

1) If the data under statement I alone is sufficient to answer the question.

2) If the data under statement II alone is sufficient to answer the question.

3) If the data under both the statements (I and II) together are necessary to answer the question.

4) If the question cannot be answered even by using both the statements together.

20) What is the difference between S.I and C.I?

600 5% 2y
1st year = 60
2nd year = 60 + 3
C.I - S.I = 3

I) The sum at 5% p.a and 2 years

II) The sum is Rs. 600



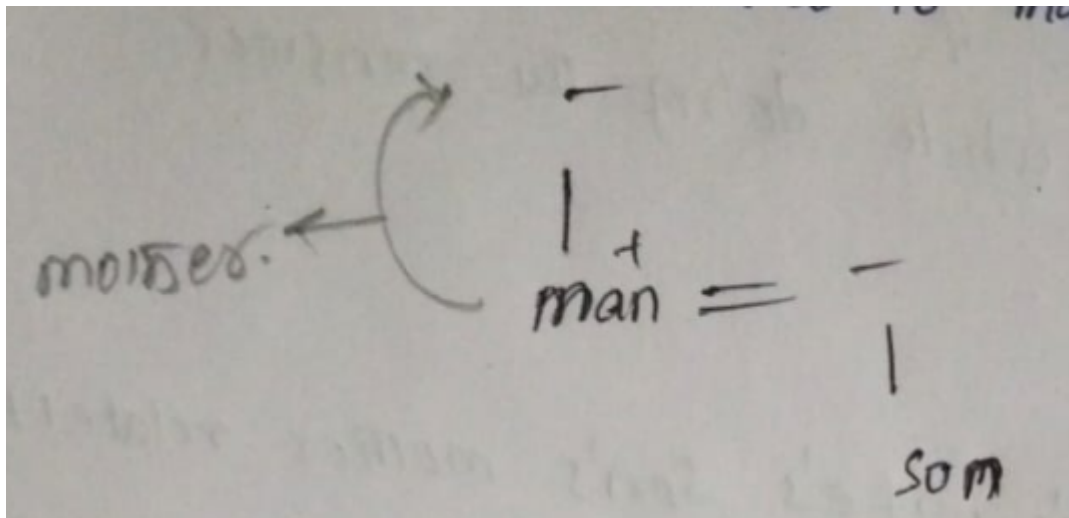
- 1) If the data under statement I alone is sufficient to answer the question.
- 2) If the data under statement II alone is sufficient to answer the question.
- 3) If the data under both the statements (I and II) together are necessary to answer the question.
- 4) If the question cannot be answered even by using both the statements together.

MODULE 2

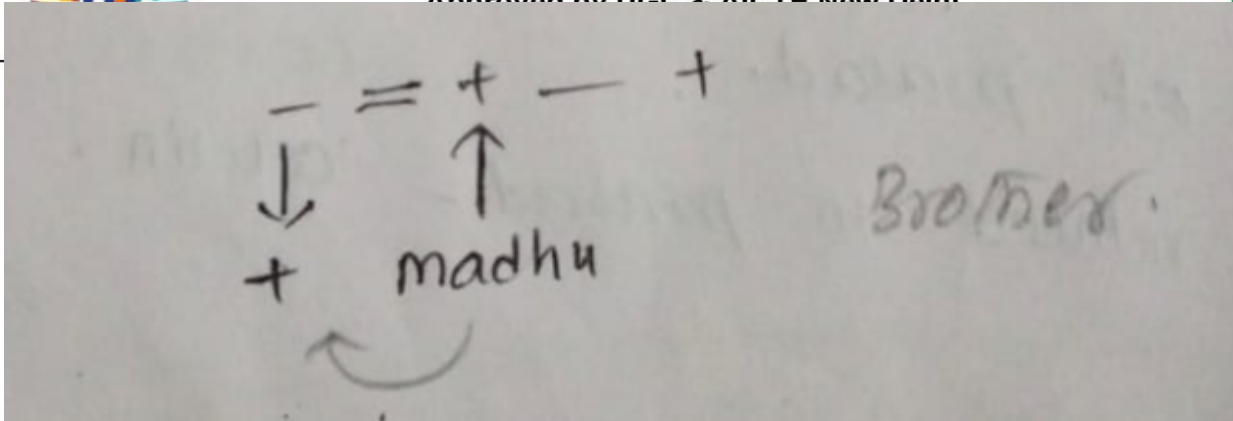
Blood Relation

Short Answers

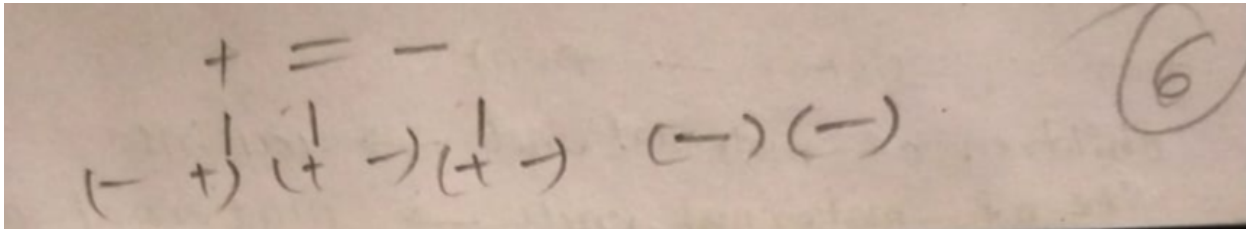
1) Pointing to a woman in a photograph a man Says" she is the mother of the father of my wife's Only Son" How is the Women related to man?



2) Pointing to a Joker, Madhu Says "He is the Son of my father's brother Only Sister-in-law" How Joker is related to Madhu?



3) A family consists of a husband and wife, their three sons and two daughters, three wives of three sons. How many females are in the family?

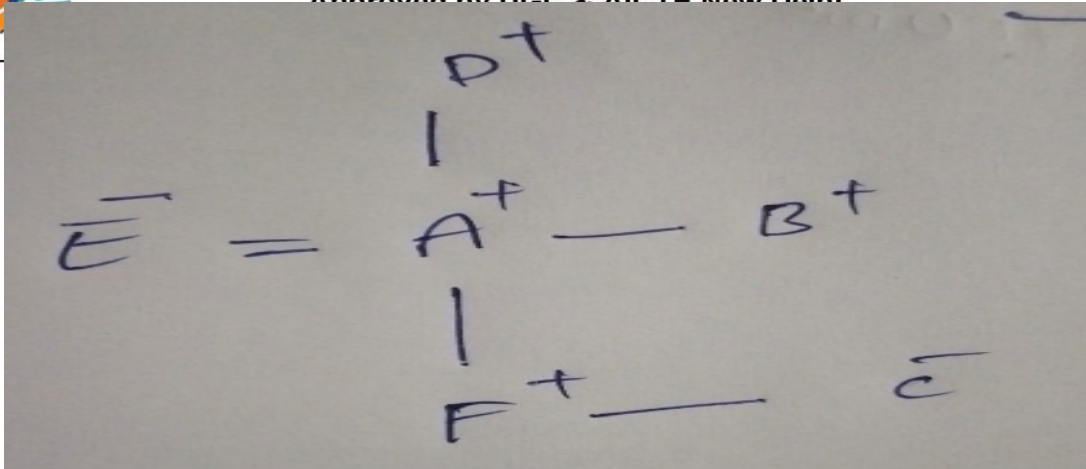


MODULE 2

Blood Relation

Long Answers

1) Study the following information carefully and answer the questions given below it. There are six persons A, B, C, D, E and F. C is the sister of F. B is the brother of E's husband. D is the father of A and grandfather of F. There are two fathers, three brothers and a mother in the group.



1. Who is the mother?

- (a) A (b) B (c) D (d) E

2. Who is E's husband?

- (a) B (b) C (c) A (d) F

3. How many male members are there in the group?

- (a) One (b) Two (c) Three (d) Four

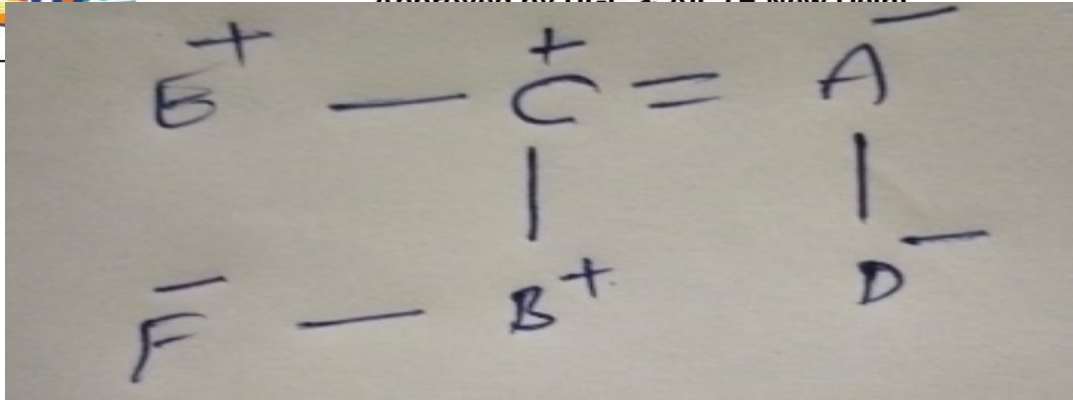
4. How is F related to E?

- (a) Uncle (b) Husband (c) Son (d) Daughter

5. Which of the following is a group of brothers?

- (a) ABF (b) ABD (c) BFC (d) BDF

2) Study the following Information and answer the questions given below it: All the six members of a family A, B, C, D, E and F are travelling together. B is the son of C but C is not the mother of B. A and C are a married couple. E is the brother of C. D is the daughter of A. F is the brother of B.



1. How many male members are there in the family?

- (a) 1** **(b) 2** **(c) 3** **(d) 4**

2. Who is the mother of B?

- (a) D** **(b) F** **(c) E** **(d) A**

3. How many children does A have?

- (a) One** **(b) Two** **(c) Three** **(d) Four**

4. Who is the wife of E?

- (a) A** **(b) F** **(c) B** **(d) Cannot be determined**

5. Which of the following is a pair of females?

- (a) AE** **(b) BD** **(c) DF** **(d) AD**

3. Study the following statements and answer the questions given below:

- (i) ' $P \div Q$ ' means 'P is son of Q'. (ii) ' $P \times Q$ ' means 'P is sister of Q'.
(iii) ' $P + Q$ ' means 'P is brother of Q'. (iv) ' $P - Q$ ' means 'P is mother of Q'.

1. How is T related to S in the expression 'T × R + V ÷ S'?

- (A) Sister** **(B) Mother** **(C) Aunt** **(D) None of these**

2. How is T related to S in the expression ' $T \times R - V + S$ '?

- (A) Father** **(B) Sister** **(C) Uncle** **(D) Aunt**



3. How is T related to S in the expression 'T + R - V + S'?

- (A) Maternal Uncle (B) Nephew (C) Son (D) Can't be determined

4. Which of the following means that 'S is the husband of T'?

- (A) $T \times R - V + S$ (B) $T - R \div V \times S$ (C) $T - R + V \div S$ (D) $T \div R \times V + S$

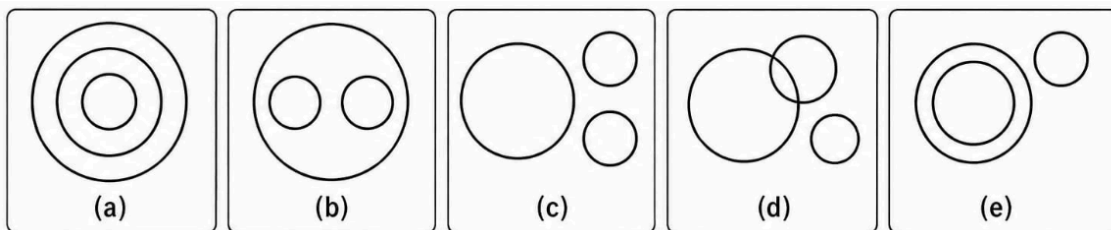
5. How is V related to T in the expression ' $T \div R + V \times S$ '?

- (A) Cousin (B) Nephew (C) Niece (D) Aunt

MODULE 3 Venn Diagrams

Short Answers

1) DIRECTIONS (Questions 1 to 2): Each of the questions below contains three elements. These three elements may or may not have some linkage. Each group of the elements may fit into one of the diagrams at (a), (b), (c), (d) and (e). You have to indicate groups of elements in each of the questions which fit which into which of the diagrams given below. The letter indicating the diagram is the answer.

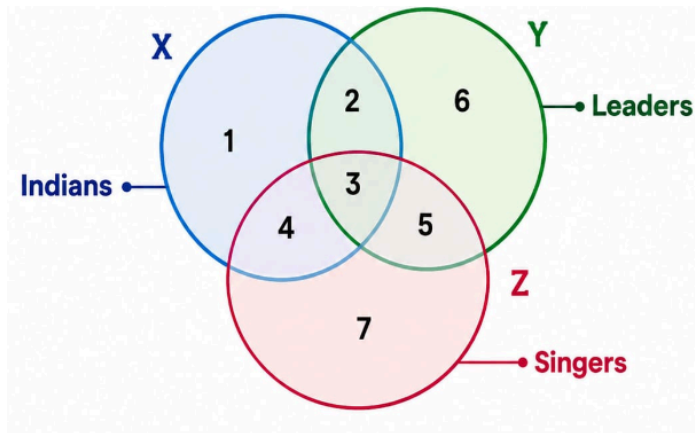


1. Vegetables, Potato, Cabbage (B)

2. Table, Chair, Furniture (B)



2) Directions (Questions 1 to 2): Study the following figure carefully and answer the given questions.



1) . Which region denotes Indian leaders who are not singers?

(a) 2

(b) 3

(c) 4

(d) 5

2) Which region represents Indian leaders who are singers?

(a) 2

(b) 3

(c) 4

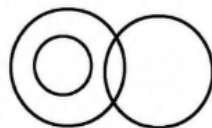
(d) 5

3. Here four diagrams (A), (B) and (D) are given. There are three elements in each the questions. Choose the Venn-diagram best illustrates the relationship among the given elements.

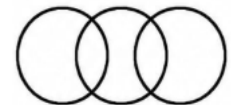
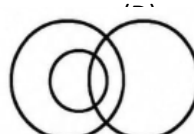
(A)



(B)



(C)



1. Sun, Stars, Moon (A)

2. Female, Mothers, Children (C)

MODULE 3

Venn Diagrams

Long Answers



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1) In an examination 35% of the students failed in Math, 25% of the students failed in English, 45% of the students failed in Hindi. If 10% of the students failed in Math and English, 20% of the students failed in English and Hindi, 30% of the students failed in Math and Hindi and 5% of the students failed in all the three subjects Math, English and Hindi.

1. What percentage of students failed in at least one subject?

- A) 40% B) 45% C) 50% D) 55%

2. What percentage of students passed in all the three subjects?

- A) 50% B) 45% C) 55% D) 60%

3. What percentage of students failed only in Mathematics?

- A) 0% B) 5% C) 10% D) 15%

4. What percentage of students failed only in English?

- A) 5% B) 0% C) 10% D) 15%

5. What percentage of students failed only in Hindi?

- A) 0% B) 5% C) 10% D) 15%

2) In an examination 40% of the students failed in Math, 30% of the students failed in English, 50% of the students failed in Hindi. If 25% of the students failed in Math and English, 15% of the students failed in English and Hindi, 22% of the students failed in Math and Hindi and 13% of the students failed in all the three subjects Math, English and Hindi.

1. What percentage of students failed in at least one subject?

- A) 69% B) 71% C) 73% D) 75%

2. What percentage of students passed in all the three subjects?

- A) 21% B) 25% C) 29% D) 31%

3. What percentage of students failed only in Mathematics?

- A) 3% B) 6% C) 9% D) 12%

4. What percentage of students failed only in Hindi?



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A) 22%

B) 24%

C) 38%

D) 18%

5. What percentage of students failed in exactly two subjects?

A) 20%

B) 23%

C) 24%

D) 25%

IV. There are total 700 students in a class. 200 students passed in Mathematics, 150 students in English and 300 students in Hindi. 50 students passed in Mathematics and English and 20 students passed in Hindi and Mathematics. 30 students passed in English and Hindi only. 10 students passed in all three subjects English, Hindi and Mathematics. Answer the following questions based on this information.

1. What is the number of students who have passed?

(A) 600

(B) 550

(C) 150

(D) 400

2. What is the number of students who have failed in all three subjects?

(A) 600

(B) 550

(C) 150

(D) 400

3. What is the number of students who have failed in Hindi only?

(A) 40

(B) 60

(C) 10

(D) 80

4. What is the number of students who have failed in English only?

(A) 10

(B) 20

(C) 30

(D) 40

5. What is the number of students who have failed in Mathematics and Hindi only?

(A) 30

(B) 70

(C) 10

(D) 40



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